higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 4

(6050364)

27 February 2019 (Y-Paper)
13:00–16:00

Scientific calculators may be used.

This question paper consists of 5 pages and 1 periodic table.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use 3,14 for the value of π
5. Start each section on a NEW page.
6. Write neatly and legibly.

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QUESTION 1

1.1 Define the following terms:

1.1.1 *Concentrate* (2)

1.1..2 *Decantation* (2)

1.2 Describe the function of the following in the absorption process:

1.2.1 Wash towers and scrubbers. (2)

1.2.2 Tray columns. (2)

1.3 List THREE advantages of the packed column technique. (3)

1.4 Discuss the solvent extraction processes as used in the absorption process. (10)

1.5 Differentiate between the construction of an agitator and a thickener as it is used in the CCD process. (4)

1.6 Fouling is always a major problem that occurs in the decantation process.

1.6.1 Explain the term fouling in decantation processes. (2)

1.6.2 How does extensive fouling affect the decantation process? (2)

1.6.3 How can fouling be remedied? (1)

[30]

QUESTION 2

2.1 Define a *closed system.* (2)

2.2 Write down the equation for each of the following shapes to calculate its volume:

2.2.1 Cone (3)

2.2.2 Cylinder (3)

2.3 State the law of conservation of mass. (3)

2.4 Define the term chemical reaction. (2)

2.5 Describe the relationship between one mole and Avogadro's number. (3)

2.6 Convert the following quantities to the unit as indicated:

2.6.1 3000 L = …….. kL (1)

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2.6.2 900 mg = ……. g (2)

2.7 Write and balance the reaction as the reactants below are added together

2.7.1 $Ca(OH)_2 + K_2CO_3 \longrightarrow$ …….. + …….. + …….. (3)

2.7.2 $Zn + 2HCl \longrightarrow$ …… + …… (2)

2.8 Calculate the atomic mass of the compounds below:

2.8.1 Na_2CO_3 (2)

2.8.2 $Al_2(CO_3)$ (2)

2.8.3 HCl (2)
[30]

QUESTION 3

3.1 Define the term *kinetic energy* of an object and write down the equation to calculate kinetic energy of an object. (4)

3.2 Convert 940 K to degrees Celsius. (4)

3.3 Calculate the amount of energy that is needed to increase the temperature of a granite object by 230 ^{0}F. The heat capacity of granite is taken as 0.79 (8)

3.4 Differentiate between solids and liquids in terms of the position and movement of their particles as the material go through the four different phases. (4)

3.5 Give the name of the phase transition during the heating and cooling of materials cross the different phases for each of the following scenarios:

3.5.1 Temperature at which a substance changes from a liquid to gas at atmospheric pressure. (2)

3.5.2 Temperature at which a substance changes from solid to liquid at atmospheric pressure. (2)

3.5.3 Temperature at which a substance changes from gas to liquid at atmospheric pressure. (2)

3.6 Define the term radiation and explain how the sun radiates electromagnetic waves in order to heat our planet. (4)
[30]

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QUESTION 4

4.1 List FOUR applications of plastics. (4)

4.2 Differentiate between cellulose and lignin as it is found in natural composites like wood. (4)

4.3 Name TWO individual processes that forging consists of. (2)

[10]

TOTAL : [100]

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THE PERIODIC TABLE OF ELEMENTS
DIE PERIODIEKE TABEL VAN ELEMENTE

1 (I)	2 (II)	3	4	5	6	7	8	9	10	11	12	13 (III)	14 (IV)	15 (V)	16 (VI)	17 (VII)	18 (VIII)
																	2 He 4
1 H 1 2,1												5 B 11 2,0	6 C 12 2,5	7 N 14 3,0	8 O 16 3,5	9 F 19 4,0	10 Ne 20
3 Li 7 1,0	4 Be 9 1,5											13 Aℓ 27 1,5	14 Si 28 1,8	15 P 31 2,1	16 S 32 2,5	17 Cℓ 35,5 3,0	18 Ar 40
11 Na 23 0,9	12 Mg 24 1,2											31 Ga 70 1,6	32 Ge 73 1,8	33 As 75 2,0	34 Se 79 2,4	35 Br 80 2,8	36 Kr 84
19 K 39 0,8	20 Ca 40 1,0	21 Sc 45 1,3	22 Ti 48 1,5	23 V 51 1,6	24 Cr 52 1,6	25 Mn 55 1,5	26 Fe 56 1,8	27 Co 59 1,8	28 Ni 59 1,8	29 Cu 63,5 1,9	30 Zn 65 1,6	49 In 115 1,7	50 Sn 119 1,8	51 Sb 122 1,9	52 Te 128 2,1	53 I 127 2,5	54 Xe 131
37 Rb 86 0,8	38 Sr 88 1,0	39 Y 89 1,2	40 Zr 91 1,4	41 Nb 92 1,6	42 Mo 96 1,8	43 Tc 1,9	44 Ru 101 2,2	45 Rh 103 2,2	46 Pd 106 2,2	47 Ag 108 1,9	48 Cd 112 1,7	81 Tℓ 204 1,8	82 Pb 207 1,8	83 Bi 209 1,9	84 Po 2,0	85 At 2,5	86 Rn
55 Cs 133 0,7	56 Ba 137 0,9	57 La 139 1,1	72 Hf 179 1,6	73 Ta 181	74 W 184	75 Re 186	76 Os 190	77 Ir 192	78 Pt 195	79 Au 197	80 Hg 201						
87 Fr 0,7	88 Ra 226 0,9	89 Ac															

58 Ce 140	59 Pr 141	60 Nd 144	61 Pm	62 Sm 150	63 Eu 152	64 Gd 157	65 Tb 159	66 Dy 163	67 Ho 165	68 Er 167	69 Tm 169	70 Yb 173	71 Lu 175
90 Th 232	91 Pa	92 U 238	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

KEY / *SLEUTEL*

Atomic number *Atoomgetal*

Electronegativity *Elektronegatiwiteit*

Symbol *Simbool*

Approximate relative atomic mass *Benaderde relatiewe atoommassa*

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